LocalResponseNorm#

https://pytorch.org/docs/stable/generated/torch.nn.LocalResponseNorm.html

Description:

Applies local response normalization over an input signal. The input signal is composed of several input planes, where channels occupy the second dimension. Applies normalization across channels. size (int) – amount of neighbouring channels used for normalization alpha (float) – multiplicative factor. Default: 0.0001 beta (float) – exponent. Default: 0.75

Function Signature

```
class torch.nn.LocalResponseNorm(size, alpha=0.0001,
beta=0.75, k=1.0)[source]#

Parameters

Shape:

extra_repr()[source]#

forward(input)[source]#
```

Returns:

Tensor

Code Examples

```
>>> lrn = nn.LocalResponseNorm(2)
>>> signal_2d = torch.randn(32, 5, 24, 24)
>>> signal_4d = torch.randn(16, 5, 7, 7, 7, 7)
>>> output_2d = lrn(signal_2d)
>>> output_4d = lrn(signal_4d)
```

Detailed Documentation

Documentation

Applies local response normalization over an input signal.

The input signal is composed of several input planes, where channels occupy the second dimension. Applies normalization across channels.

size (int) – amount of neighbouring channels used for normalization

alpha (float) – multiplicative factor. Default: 0.0001

beta (float) – exponent. Default: 0.75

k (float) – additive factor. Default: 1

Input: $(N,C,*)$(N, C, *)$(N,C,*)$

Output: $(N,C,*)$(N, C, *)$(N,C,*)$ (same shape as input)

Return the extra representation of the module.

Runs the forward pass.